DS - reference sheet.

$$\sum_{k\geq 0} \frac{(-1)^k}{(2k)!}\beta^{2k} = \cos\beta$$
$$\sum_{k\geq 0} \frac{(-1)^k}{(2k+1)!}\beta^{2k+1} = \sin\beta$$

- real part > 0 => trigon.
- real part < 0 => clock.

CONTINUOUS DYNAMIC SYSTEMS

↳ differential equations. (t = time = continuous variable)

DISCRETE DYNAMIC SYSTEMS

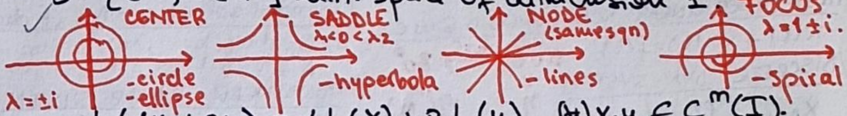
↳ difference equations. (k = discrete variable)

$S = \{c_1 \cdot t + c_2 \cdot 1 ; c_1, c_2 \in \mathbb{R}\}$ lin. space of dimension 2.

$S = \{c \cdot 1, c \in \mathbb{R}\}$ lin. space of dimension 1.

$S = \{\cdots\}$ = subspace of $C(\mathbb{R})$.

$C(\mathbb{R})$ = the set of all cont. fct.



LINEAR MAP:

$f(k_1 v_1 + k_2 v_2) \overset{?}{=} k_1 \cdot f(v_1) + k_2 \cdot f(v_2)$

$L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)$, $(\forall)\, x, y \in C^m(I)$. $(\forall)\, \alpha, \beta \in \mathbb{R}$.

SUBSPACE:

$S \leq_K V$ $\begin{cases} S \neq \emptyset \\ (\forall) k_1, k_2 \in K \\ (\forall) v_1, v_2 \in S \end{cases}$, $k_1 v_1 + k_2 v_2 \in S$.

$$\boxed{ch\, t = \frac{1}{2} \cdot e^t + \frac{1}{2} \cdot e^{-t}}$$

$$\boxed{sh\, t = \frac{1}{2} \cdot e^t - \frac{1}{2} \cdot e^{-t}}$$

$(ch\, t)' = sh\, t$

$(sh\, t)' = ch\, t$

$ch\, 0 = 1$

$sh\, 0 = 0$

$ch^2 t - sh^2 t = 1$.

FIRST-ORDER SCALAR D.E: $x'(t) + a(t) \cdot x(t) = b(t)$ (non-homog.)

=> integrating factor.

FIRST-ORDER SCALAR AUTONOM. D.E. $x'(t) + a(t)x(t) = 0$ (homog.)

$x' = f(x)$; $x' = x^2 - 5x + 6 \to$ sol: $\varphi'(t) = f(\varphi(t))$

=> equil., phase portrait, flow, attr., rep, long-term beh.

=> sign table; $\eta^* \in \mathbb{R}$ not equil => "regular point".

LINEAR PLANAR SYS.

$x' = Ax$

$\begin{cases} x' = a_{11}x + a_{12}y \\ y' = a_{21}x + a_{22}y \end{cases}$

=> eigenvalues

=> eigenvectors...

SECOND-ORDER LINEAR D.E. w C.C: $x'' + ax' + bx = 0$ (homog.)

=> char. eq. (homog.)

=> $x = x_h + x_p$ (non-homog.) $x'' + ax' + bx = g(t)$ (non-homog.)

LINEAR NON-HOMOG. PLANAR SYSTEM

$x' = Ax + b$

DIFFERENCE EQUATION (DISCRETE SCALAR SYSTEM)

$x_{k+1} = f(x_k)$; $x_{k+1} = x_k + \lambda x_k (2 - x_k)$

$|f'(x^*)| < 1$ => attractor ; $|f'(x^*)|$ => repeller.

GENERAL NON-LINEAR PLANAR SYS.

$\begin{cases} x' = f(x,y) \\ y' = g(x,y) \end{cases}$ => equil => $J(x^*, y^*)$ =>

eigenvalues determine stability only if hyperbolic

LINEAR DIFFERENCE EQUATION : $x_{k+1} + a x_k = b$

=> char. eq. (first-order linear non-hom. d.e.)

"ASSOCIATED" PLANAR SYSTEM.

$x'' = F(x, x') \to$ let $y = x'$ =>...

ex: $x'' + 9x = 0$ => $\begin{cases} x' = y \\ y' = -9x \end{cases}$

SECOND-ORDER-DIFFERENCE EQUATION

$x_{k+2} + a \cdot x_{k+1} + b \cdot x_k = 0$ => char. eq.

NON-LINEAR PLANAR SYS: ex: $\begin{cases} x' = y \\ y' = -9\sin x \end{cases}$, $\begin{cases} x' = x(1-y) \\ y' = y(2-x) \end{cases}$

LHDE: $\boxed{x^{(m)} + a_1(t) \cdot x^{(m-1)} + \ldots + a_{m-1}(t) \cdot x' + a_m(t) \cdot x = 0}$ (1)

m = ORDER ; $a_1, \ldots, a_m \to$ coefficients. (contin. fct.)

Eigenvalue = $\lambda \in \mathbb{C}$ s.t. $\exists v \neq 0$ s.t. $Av = \lambda v$.

$\det(A - \lambda I_m) = 0$ char. polyn.

$x' = Ax$; If $Re(\lambda) < 0$ => $\lim_{t \to \infty} e^{At} = 0$

POLAR-COORD. NON-LIN. SYS

$\begin{cases} x' = -y + \ldots \\ y' = x + \ldots \end{cases}$ $\rho^2 = x^2 + y^2$ => $\rho', \dot{\theta}$?

$tg\theta = \frac{y}{x}$

$x = \rho\cos\theta$, $y = \rho\sin\theta$

IVP: $\begin{cases} \text{eq. (1)} \\ x(t_0) = \eta_1, x'(t_0) = \eta_2, \ldots, x^{(m-1)}(t_0) = \eta_m. \end{cases}$

E and U theorem for LHDE:

IVP has unique sol.

Exponential matrix:

$e^a = \sum_{k=0}^{\infty} \frac{1}{k!} a^k$ abs. convergent

$A \in M_2(\mathbb{R})$

$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k$ abs. convergent,

$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$

$e^{\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} \to \begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix}$

$e^{tA} = P \cdot e^{tB} \cdot P^{-1}$

P invertible, $A = P \cdot B \cdot P^{-1}$

NEWTON ITERATION AS A DISCRETE DYN. SYSTEM

$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)}$, $g(x) = 0$

=> $f'(x)$ ....

COBWEB DIAGRAM FOR SCALAR MAPS

$g(x) = x - \frac{f(x)}{f'(x)}$ → plot and ∩ with $y = x$

(basin of attraction)

Solution for LHDE:

a function $\varphi : I \to \mathbb{R}$

s.t. $\begin{cases} \varphi \in C^m(I) \\ \varphi^{(m)}(t) + a_1(t) \cdot \varphi^{(m-1)}(t) + \ldots + a_m(t) \cdot \varphi(t) = 0, (\forall) t \in I. \end{cases}$

Principal matrix sol for $x' = Ax$

$\begin{cases} E'(t) = A \cdot E(t) \\ E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{cases}$; $E : \mathbb{R} \to M_2(\mathbb{R})$

UNCOUPLED SYSTEM: ($a_{12} = a_{21} = 0$)

$\begin{cases} x' = a_{11} \cdot x + 0 \cdot y \\ y' = 0 \cdot x + a_{22} \cdot y \end{cases}$

Fundamental theorem for LHDE:

the set of all solutions of (1) is a LINEAR space of dim = m.

$D^2(x) = D(D(x)) = D(x') = x''$

diagonal matrix <=> elems outside DP = 0

General sol: $\begin{cases} x = c_1 \cdot e^{a_{11} \cdot t} \\ y = c_2 \cdot e^{a_{22} \cdot t} \end{cases}$

Euler's formula:

$e^{\alpha + i\beta} = e^{\alpha} \cdot e^{i\beta} = e^{\alpha} \cdot (\cos\beta + i\sin\beta)$

IVP $\Leftrightarrow$ exactly m initial conditions for an mth order diff. eq. ✓

~~Analize strengths~~

SECOND-ORDER LINEAR D.E w non-ct coeff.

$x'' + t^2 \cdot x = 0 \Rightarrow \exists!$

$\rightarrow$ ~~Lin. sol. ...~~

✓**FIRST-ORDER LINEAR HOMOG. D.E. w c.c.**

$x'(t) + \lambda x(t) = 0, \ \lambda \in \mathbb{R}$.

✓**FIRST-ORDER LINEAR NON-HOMOG. D.E w c.c.**

$x'(t) + \lambda x(t) = f(t)$

$100°C \xrightarrow{t=5} 80°$

$\eta \qquad \varphi(t,\eta)$

✓**DISCRETE DYN. SYSTEM**

$x_{m+1} = f(x_m)$ $\quad \varphi(t,\eta) = (100-20)e^{-5t} + 20 \Rightarrow 80$ ...

Lin. theorem for discrete dyn. sys:

$|f'(\eta^*)| < 1 \Rightarrow$ attractor

$|f'(\eta^*)| > 1 \Rightarrow$ ~~repeller~~ / unstable — repeller / unstable

$f'(\text{fixed point}) > 1$

✓**PRINCIPAL MATRIX SOLUTION $E(t)$**

$E(0) = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

✓**FIRST-ORDER LIN. HOMOG. SYS. W C.C.**

$\dot{x} = Ax, \ x(0) = \eta$ (IVP)

$\rightarrow$ unique sol: $x(t) = e^{tA} \cdot \eta$

$\rightarrow$ flow: $\varphi(t,\eta) = e^{tA} \cdot \eta$

$\rightarrow$ if $\det A \neq 0 \Rightarrow$ unique eq: $x = 0$

$\rightarrow$ fundam. matrix: $\Phi(t) = e^{tA}$

? General sol of $x' = Ax$, A diag. over $\mathbb{C}$ $\Rightarrow$ eigenvalues $\Rightarrow$ eigenvectors

$\Rightarrow D = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \ P = (u_1, \dots, u_n) \Rightarrow x(t) = P \cdot \operatorname{diag}(e^{t\lambda_1} \dots) P^{-1} \cdot c$

$A = PDP^{-1} \Rightarrow e^{tA} = P \cdot e^{tD} \cdot P^{-1}$.

✓**NONLINEAR PLANAR SYSTEM**

$\begin{cases} \dot{x} = f_1(x,y) \\ \dot{y} = f_2(x,y) \end{cases} \Leftrightarrow \dot{x} = f(x)$

✓Basin of attraction of η^* = attractor:

$A\eta^* = \{\eta \in \mathbb{R} : \lim_{t \to \infty} \varphi(t,\eta) = \eta^*\}$

✓Basin of attr. for discrete systems:

$A\eta^* = \{\eta : \lim_{m \to \infty} x_m = \eta^*\}$

✓$\Rightarrow$ ~~IVP~~ A solution is a function $\varphi: I \to \mathbb{R}$

I = interval, $\varphi \in C^1$, $\varphi'(x) = g(x, \varphi(x))$

$\rightarrow$ if $\operatorname{Re}(\lambda) < 0 \Rightarrow \lim_{t \to \infty} e^{At} = 0. \Leftrightarrow$

$\Leftrightarrow \lim_{t \to \infty} \varphi(t,\eta) = 0$.

✓$e^{it} = (\cos t + i \sin t)$

✓$e^{-it} = (\cos t - i \sin t)$

✓Matrix exponentials:

$e^{tA} = I + t \cdot A + \frac{t^2 A^2}{2!} + \dots$

? **PHASE PORTRAIT FOR COMPLEX SOL.**

$z(t) = e^{(\alpha + i\beta)t} z_0$

$|z(t)| = |z_0| \cdot e^{\alpha t}$

$\arg(z(t)) = \arg(z_0) + \beta t$

✓**Euler numerical:**

$(h = ct) \begin{cases} x_{k+1} = x_k + h, \ k = \overline{0, n-1} \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{cases}$ $\quad h = \frac{x^* - x_0}{m}$

✓**LIN. METHOD FOR A NONLINEAR PLANAR SYS**

Linearised system: $\dot{x} = Jf(x^*)(x - x^*)$

x^* = equil.

✓NON-HYP. EQUIL $\Rightarrow$ linearization fails $\Rightarrow$ use first integral for proving stability

$y' = g(x,y) \Rightarrow$ IVP: $\begin{cases} y' = g(x, \varphi(x)) \\ y(x_0) = y_0 \end{cases}$

! DEFINITION: ?

✓**FIRST INTEGRALS**: $H: D \subseteq \mathbb{R}^n \to \mathbb{R}$, $H(\varphi(t,\eta)) = H(\eta)$

• H = f.i. $\Leftrightarrow H(\varphi(t,\eta)) = H(\eta)$; $\nabla H(x,y) \neq 0 \Rightarrow H$ not locally ct.

• a global attractor does not have H; first int. are not locally ct.

✓**GLOBAL ATTRACTOR** $(= \eta^*)$

$\Rightarrow \lim_{t \to \infty} \varphi(t,\eta) = \eta^*$, $(\forall) \eta \in \mathbb{R}^2$.

✓**GLOBAL FIRST INTEGRALS**: $H: \mathbb{R}^m \xrightarrow{\text{(global)}} \mathbb{R}$

$\boxed{\eta^* \text{ equilibrium} \Rightarrow \varphi(t, \eta^*) = \eta^*}$

• defined on all $\mathbb{R}^m$

• $H \in C^1(\mathbb{R}^m)$

• H not locally ct. $\rightarrow \nabla H = 0$ $\rightarrow$ (x,y) critical point $\Rightarrow$ not locally ct. / $\rightarrow (\forall)$ points (ALL) $\Rightarrow$ locally ct.

✓• $H(\varphi(t,x)) = H(x)$ or $\nabla H \cdot f = 0$

$\boxed{H \text{ global f.i.} \Leftrightarrow \frac{\partial H}{\partial x}(x,y)(\dot{x}) + \frac{\partial H}{\partial y}(x,y)(\dot{y}) = 0}$

("first-order partial diff. eq") (*)

✓$\lambda = a \pm bi \Rightarrow \begin{cases} x(t) = e^{at} \cdot \sin(bt) \\ y(t) = e^{at} \cdot \cos(bt) \end{cases}$

✓**ASSOCIATED PLANAR SYSTEM**

$\theta'' + w^2 \theta = 0 \Rightarrow x = \theta, \ y = \theta'$

$\Rightarrow \begin{cases} x' = y \\ y' = -w^2 x \end{cases}$

✓**BARBALAT**: $\varphi: \mathbb{R} \to \mathbb{R}$, C^1 fct. st. $\exists \lim_{t \to \infty} \varphi(t)$, $\lim_{t \to \infty} \varphi'(t)$ finite $\Rightarrow \lim_{t \to \infty} \varphi'(t) = 0$

Exam templates:

second order ODE w. c.c.

$x'' - \sqrt{3}x' + x = 0$

$r^2 - \sqrt{3}r + 1 = 0$

$\Rightarrow r_1 \neq r_2 \in \mathbb{R} \Rightarrow x = c_1 \cdot e^{r_1 t} + c_2 \cdot e^{r_2 t}$

$\Rightarrow r_1 = r_2 \in \mathbb{R} \Rightarrow x = (c_1 + c_2 \cdot t) e^{rt}$

$\Rightarrow r \in \mathbb{C} \Rightarrow x = e^{\alpha t}(c_1 \cos \beta t + c_2 \sin \beta t)$

second order d.e

$x_{k+2} - \sqrt{3}x_{k+1} + x_k = 0$

$r^2 - \sqrt{3}r + 1 = 0$

$\Rightarrow r_1 \neq r_2 \in \mathbb{R} \Rightarrow x_k = c_1 \cdot r_1^k + c_2 \cdot r_2^k$

$\Rightarrow r_1 = r_2 \in \mathbb{R} \Rightarrow x_k = (c_1 + c_2 \cdot k) r^k$

$\Rightarrow r \in \mathbb{C} \Rightarrow r = \rho e^{i\theta} \Rightarrow x_k = \rho^k \cdot (c_1 \cos(k\theta) + c_2 \cdot \sin(k\theta))$

✓**JACOBIAN**

$\uparrow$

ONLY FOR NON-LINEAR

✓φ sol. of $\dot{x} = f(x) \Rightarrow \varphi'(t) = f(\varphi(t))$, $(\forall) t \in \mathbb{R}$

✓non-linear planar sys.

Lin: $\dot{x} = -y + x \ (\dots)$

$\dot{y} = x + y \ (\dots)$

$\Rightarrow$ equil, Jacobian

$\Rightarrow$ eigenvals $\Rightarrow$ classify.

hyp $\Leftrightarrow \operatorname{Re}(\lambda) \neq 0$

✓phase portrait from f.i.

$\Rightarrow$ equil $\Rightarrow H(x,y) = c \Rightarrow$

$\Rightarrow$ direction using x', y'

$\Rightarrow$ arrows

ex: $H = x^2 + y^2 \Rightarrow$ level curves = circles

$\Rightarrow$ sign table (take a point $\in$ curve $\Rightarrow$ $\Rightarrow$ compute x', y'; if $> 0 \Rightarrow$ up.

✓If $f \in C^1 \Rightarrow$ IVP has a unique maximal sol.

✓**first integrals**

$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}} \Rightarrow$ separate var $\Rightarrow$ integrate $\Rightarrow$ verify using PDE *

Euler stability

"attractor preserved": $x' = f(x)$; $T(x) = x + h \cdot f(x)$

$\Rightarrow$ attractor equil. $(x^*) \Rightarrow |T'(x^*)| < 1 \Rightarrow$ attr.

$\Rightarrow$ solve for h.

✓**lin. system $\Rightarrow$ second-order ODE**

$\begin{cases} \dot{x} = \dots \\ \dot{y} = \dots \end{cases}$ $\overset{?}{\Rightarrow} \ddot{x} = \dots$

✓**linear scalar ODE**

$x' + 3x = 2$

$\mu(t) = e^{\int a(t)\,dt}$

$x' + 3x = 2 \ | \cdot \mu(t)$

(int. factor)

Euler formula

$\begin{cases} x_k = k \cdot h \\ y_{k+1} = y_k + h \cdot y_k \end{cases}$

ex: $x' = f(x)$

$x_{m+1} = x_m + h \cdot f(x_m)$

✓**Scalar autonomous eq (equil + stab.)**

$x' = f(x)$

$f(x) = 0 \Rightarrow x^* \dots$

Lin: $f'(x^*) < 0 \Rightarrow$ attr.

$f'(x^*) > 0 \Rightarrow$ rep.

$f'(x^*) = 0 \Rightarrow$?

✓**lin. systems (type + stab.)**

$x' = Ax \Rightarrow$ eigenvals $\Rightarrow$ classif.

solve system

$x' = Ax \Rightarrow$ eigenvals $\Rightarrow$ eigenvect.

$\Rightarrow x(t) = c_1 \cdot e^{\lambda_1 t} \cdot v_1 + c_2 \cdot e^{\lambda_2 t} \cdot v_2 \dots$

✓**when linearization fails**: $(\operatorname{Re}(\lambda) = 0)$.

ex: $\begin{cases} \dot{x} = y \\ \dot{y} = -9 \sin x \end{cases}$ $\Rightarrow$ try lin: $J(x,y) = \begin{pmatrix} 0 & 1 \\ -9\cos x & 0 \end{pmatrix} \Rightarrow J(0,0) = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$

$\Rightarrow \lambda^2 + 9 = 0 \Rightarrow \lambda = \pm 3i \Rightarrow \operatorname{Re}(\lambda) = 0 \Rightarrow$ lin. fails

$\Rightarrow$ use $H \Rightarrow \dots \Rightarrow H(x,y) = \frac{y^2}{2} + 9\underbrace{(1 - \cos x)}_{\geq 0} \geq 0$

$H(x,y) = 0 \Leftrightarrow (x,y) = (0,0)$

$\Rightarrow (0,0)$ STRICT LOCAL MIN.

$H(x,y)$ = ct $\Rightarrow$ stable (closed orbits, centers)

✓whenever the equation is LINEAR + NON-HOMOGENEOUS $\Rightarrow$ do $x = x_h + x_p$